Chapter 2

Number Sequence and Series

A number series is a sequence of numbers written from left to right in a certain pattern. To solve the questions on series, we have to detect/find the pattern that is followed in the series between the consecutive terms, so that the wrong/missing term can be find out.

Types of Series

There can be following types of series

2.1 Prime Number Series

The number which is divisible by 1 and itself, is called a prime number.
The series formed by using prime number is called prime number series.
Ex. 1 Find out the next term in the series 7,11,13,17,19,. . .

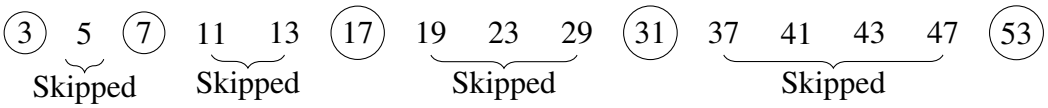
Sol. Given series is a consecutive prime number series.
Therefore, the next term will be 23.

Ex. 2 Find out the next term in the series 3, 7,17, 31,. . .

Sol. Here, every next prime number takes place skipping one, two, three and four prime numbers, respectively.

Hence, the required answer will be 53.

Let us see



2.2 Addition Series

The series in which next term is obtained by adding a specific number to the previous term, is known as addition series. Addition series are increasing order series and difference between consecutive term is equal.

Ex. 3 Find out the missing term in the series $2, 6, 10, 14, \ldots, 22$

Sol. Here, every next term is obtained by adding 4 to the previous term.

$\therefore$ Required term = 14 + 4 = 18

2.3 Difference Series

Difference series are decreasing order series in which next term is obtained by subtracting a fixed/specific number from the previous term.

Ex. 4 Find out the missing term in the series $108, 99, 90, 81, \ldots, 63$.

Sol. Here, every next number is 9 less than the previous number.

So, required number $= 81 - 9 = 72$

2.4 Multiplication Series

When each term of a series is obtained by multiplying a number with the previous term, then the series is called a multiplication series.

Note: Number which is multiplied to consecutive terms, can be fixed or variable.
Ex. 5 Find out the missing term in the series $4, 8, 16, 32, 64, \ldots, 256$.

Sol. Here, every next number is double the previous number.

So, required number $= 64 \times 2 = 128$.

Ex. 6 Find out the missing term in the series $5, 20, 80, 320, \ldots, 5120$.

Sol. Here, every next number is 4 times the previous number.

So, required number $= 320 \times 4 = 1280$

2.5 Division Series

Division series are those in which the next term is obtained by dividing the previous term by a number.
Note: Number which divides consecutive terms can be fixed or variable.
Ex. 7 Find out the missing term in the series $10080, 1440, 240, \ldots, 12, 4$.
Sol. Series Pattern

$$\frac{10080}{7} = 1440, \frac{1440}{6} = 240, \frac{240}{5} = 48, \frac{48}{4} = 12, \frac{12}{3} = 4$$

Hence, missing term is 48.
Ex. 8 Find out the missing term in the series $512, 216, 72, \ldots, 12$.
Sol.

$$\frac{512}{2} = 216, \frac{216}{3} = 72, \frac{72}{2} = 36, \frac{36}{3} = 12$$

Hence, missing term is 36.

2.6 Square and Cube Series

❒ **n^2 Series**

When a number is multiplied with itself, then it is called as square of a number and the series formed by square of numbers is called n^2 series.
Ex. 9 Find out the missing term in the series $4, 16, 36, 64, \ldots, 144$.
Sol. This is a series of squares of consecutive even numbers. Let us see

$$2^2 = 4, \quad 4^2 = 16, \quad 6^2 = 36, \quad 8^2 = 64, \quad 10^2 = 100, \quad 12^2 = 144$$

Hence, missing term is 100.

Ex.10 Find out the missing term in the series $1, 4, 9, 16, 25, \ldots, 49$.

Sol. This is a series of squares of consecutive natural numbers. Let us see

$$1^2 = 1, \quad 2^2 = 4, \quad 3^2 = 9, \quad 4^2 = 16, \quad 5^2 = 25, \quad 6^2 = 36, \quad 7^2 = 49$$

Hence, missing term is 36.

❒ **($n^2 + 1$) Series**

If in a series each term is a sum of a square term and 1, then this series is called $(n^2 + 1)$ series.

Ex. 11 Find out the missing term in the series $10, 17, 26, 37, \ldots, 65$.

Sol. Series pattern

$3^2+1 = 10; \quad 4^2+1 = 17; \quad 5^2+1 = 26; \quad 6^2+1 = 37; \quad 7^2+1 = 50; \quad 8^2+1 = 65.$

So, required number = 50

Ex. 12 Find out the missing term in the series $122, 145, 170, 197, \ldots, 257$.

Sol. Series pattern

$11^2+1 = 122;\ 12^2+1 = 145;\ 13^2+1 = 170;\ 14^2+1 = 197;\ 15^2+1 = 226;\ 16^2+1 = 257.$

So, correct answer is 226.

❒ **($n^2 - 1$) Series**

In a series, if each term is obtained by subtracting 1 from square of a number, then such series is known as $(n^2 - 1)$ series.

Ex. 13 Find out the missing term in the series $0, 3, 8, 15, 24, \ldots, 48$.

Sol. Series pattern

$1^2-1 = 0\,;2^2-1 = 3\,;3^2-1 = 8\,;4^2-1 = 15\,;5^2-1 = 24\,;6^2-1 = 35\,;7^2-1 = 48.$

So, correct answer is 35.

Ex. 14 Find out the missing term in the series $224, 195, 168, \ldots, 120$.

Sol. Series pattern

$15^2 - 1 = 224;\ 14^2 - 1 = 195;\ 13^2 - 1 = 168;\ 12^2 - 1 = 143;\ 11^2|1 = 121.$

So, required number = $12^2 - 1 = 143$.

❒ $(n^2 + n)$ **Series**

Those series in which each term is a sum of a number with square of that number, is called as $(n^2 + n)$ series.
Ex. 15 Find out the missing term in the series $12, 20, 30, 42, \ldots, 72$.
Sol. Series pattern

3^2+3 = 12, 4^2+4 = 20, 5^2+5 = 30, 6^2+6 = 42, 7^2+7 = 56, 8^2+8 = 72.

So, required number is $7^2 + 7 = 56$.
Ex. 16 Find out the missing term in the series $420, 930, 1640, \ldots, 3660$.
Sol. Series pattern

20^2+20 = 420, 30^2+30 = 930, 40^2+40 = 1640, 50^2+50 = 2550, 60^2+60 = 3660.

So, required number is $50^2 + 50 = 2550$

❒ $(n^2 - n)$ **Series**

Series in which each term is obtained by subtracting a number from square of that number, is known as $(n^2 - n)$ series.
Ex. 17 Find out the missing term in the series $42, 30, ..., 12, 6$.
Sol. Series pattern

$7^2 - 7 = 42$, $6^2 - 6 = 30$, $5^2 - 5 = 20$, $4^2 - 4 = 12$, $3^2 - 3 = 6$

So, required number is $5^2 - 5 = 20$. Ex. 18 Find out the missing term in the series $210, 240, 272, 306, \ldots, 380$.
Sol. Series pattern

$15^2 - 15$, $16^2 - 16$, $17^2 - 17$, $18^2 - 18$, $19^2 - 19$, $20^2 - 20$.

So, required number is $19^2 - 19 = 342$

❒ n^3 **Series**

A number when multiplied with itself twice, then the resulting number is called the cube of a number and series which consist of cube of different number following a specified sequence, is called as n^3 series.
Ex. 19 Find out the missing term in the series $1, 8, 27, \ldots, 125, 216$.
Sol. Series pattern

$1^3 = 3$, $2^3 = 8$, $3^3 = 27$, $4^3 = 64$, $5^3 = 125$, $6^3 = 216$.

So, required number is $4^3 = 64$.

Ex. 20 Find out the missing term in the series 1000, 8000, 27000, 64000, 125000, . . . ,

Sol. Series pattern

$10^3 = 1000,\ 20^3 = 8000,\ 30^3 = 27000,\ 40^3 = 64000,\ 50^3 = 125000,\ 60^3 = 216000.$

So, required number = $60^3 = 216000$.

❒ $(n^3 + 1)$ **Series**

Those series in which each term is a sum of a cube of a number and 1, are known as $(n^3 + 1)$ series.

Ex. 21 Find out the missing term in the series 126, 217, 344, ..., 730.

Sol. Series pattern

$5^3 + 1 = 126,\ 6^3 + 1 = 217,\ 7^3 + 1 = 344,\ 8^3 + 1 = 513,\ 9^3 + 1 = 730.$

So, required number is $8^3 + 1 = 513$

Ex. 22 Find out the missing term in the series 1001, 8001, . . . , 64001, 125001, 216001.

Sol. Series pattern

$10^3 + 1, \quad 20^3 + 1, \quad 30^3 + 1, \quad 40^3 + 1, \quad 50^3 + 1, \quad 60^3 + 1.$

So, required number is $30^3 + 1 = 27001$

❒ $n^3 - 1$ **Series**

Series in which each term is obtained by subtracting 1 from the cube of a number, is known as $(n^3 - 1)$ series.

Ex. 23 Find out the missing term in the series 0, 7, 26, 63, 124, 215, . . . ,

Sol. Series pattern

$1^3-1 = 0,\ 2^3-1 = 7,\ 3^3-1 = 26,\ 4^3-1 = 63,\ 5^3-1 = 124,\ 6^3-1 = 215,\ 7^3-1 = 342.$

So, required number is $7^3 - 1 = 342$.

Ex. 24 Find out the missing term in the series . . . , 7999, 26999, 63999, 124999.

Sol. Series pattern

$$10^3 - 1, \quad 20^3 - 1, \quad 30^3 - 1, \quad 40^3 - 1 \quad 50^3 - 1.$$

So, required number is $10^3 - 1 = 999$.

❒ $(n^3 + n)$ **Series**

When each term of a series is a sum of a number with its cube, then the series is known as $(n^3 + n)$ series.
Ex. 25 Find out the missing term in the series $2, 10, 30, \ldots, 130, 222$.
Sol. Series pattern

$$1^3 + 1, \quad 2^3 + 2, \quad 3^3 + 3, \quad 4^3 + 4, \quad 5^3 + 5, \quad 6^3 + 6.$$

So, required number is $4^3 + 4 = 68$.
Ex. 26 Find out the missing term in the series $\ldots, 8020, 27030, 64040$.
Sol. Series pattern

$$10^3 + 10, \quad 20^3 + 20, \quad 30^3 + 30, \quad 40^3 + 40.$$

So, required number is $10^3 + 10 = 1010$.

❒ **(**$n^3 - n$**) Series**

When each term of a series is obtained by subtracting a number from its cube, then series is termed as $(n^3 - n)$ series.
Ex. 27 Find out the missing term in the series $0, 6, 24, 60, 120, \ldots,$
Sol. Series pattern

$$1^3 - 1, \quad 2^3 - 2, \quad 3^3 - 3, \quad 4^3 - 4, \quad 5^3 - 5, \quad 6^3 - 6.$$

So, required number is $6^3 - 6 = 210$.
Ex. 28 Find out the missing term in the series $\ldots, 7980, 26970, 63960, 124950$.
Sol. Series pattern

$$10^3 - 10, \quad 20^3 - 20, \quad 30^3 - 30, \quad 40^3 - 40, \quad 50^3 - 50.$$

So, required number is $10^3 - 10 = 990$.

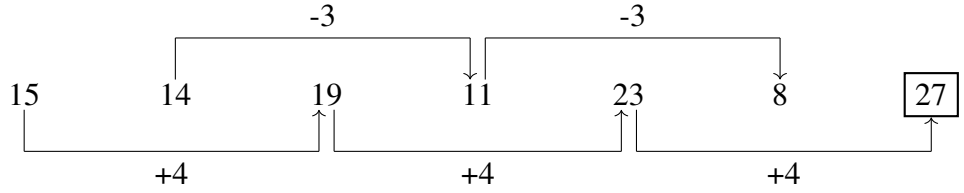
2.7 Alternating Series

In alternating series, successive terms increase and decrease alternately. The possibilities of alternating series are

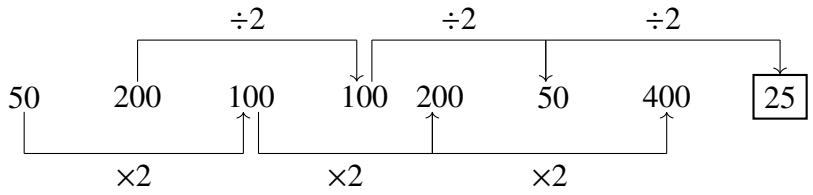
- It is a combination of two different series.

- Two different operations are performed on successive terms alternately.

Ex. 29 Find the next term in the series 15, 14, 19, 11, 23, 8, . . . ,
Sol. Series pattern is



So, the required number = 23 + 4 = 27
Ex. 30 Find the next term in the series 50, 200, 100, 100, 200, 50, 400, . . . ,
Sol. Series pattern is



So, the required number = $\frac{50}{2}$ = 25

2.8 Arithmetic Progression (AP)

The progression of the form

$$a, a + d, a + 2d, a + 3d, \ldots,$$

is known as an arithmetic progression with first term a and common difference d.
Then, nth term

$$T_n = a + (n - 1)d$$

Ex. 31 In series 359, 365, 371, . . . , what will be the 10th term?

Sol. The given series is in the form of AP.

Since, common difference *i. e.,* d is same.

Here, $a = 359$ and $d = 365 - 359 = 6$.

$\therefore$ 10th term $= a + (n - 1)d$

$= 359 + (10 - 1)6 = 359 + (9 \times 6) = 413.$

2.9 Geometric Progression (GP)

The progression of the form

$$a, ar, ar^2, ar^3, \ldots,$$

is known as a Geometric Progression (GP) with first term a and common ratio r. Then, nth term of GP

$$T_n = ar^{n-1}$$

Ex. 32 In the series $7, 14, 28, \ldots$, what will be the 10th term?

Sol. The given series is in the form of GP.

Since, common ratio i.e, r is same.

Here, $a = 7$ and $r = \dfrac{14}{7} = 2$

$\therefore$ 10th term $= ar^{n-1} = 7 \times 2^{10-1} = 7 \times 2^9 = 3584$

Types of Questions Asked on Number Series

There are mainly three types of questions which are asked from this chapter which are as follows

Type I- To Find the Missing Term

In this type of questions a series is given in which one of the term is missing and it is required to find the missing term by detecting the pattern of series.

Ex. 33 What should be the next term in the following series $12, 24, 72, 144, ?$
Sol. Series pattern is

$$12 \times 2 = 24, \quad 24 \times 3 = 72, \quad 72 \times 2 = 144, \quad 144 \times 3 = 432$$

Clearly, next term is 432.

Ex. 34 What should come in place of question mark in the series given below $92, ?, 46, 69, 138, 345$.
Sol. Series Pattern is

$$92 \times \frac{1}{2} = 46;\ 46 \times 1 = 46;\ 46 \times 1\frac{1}{2} = 69, 69 \times 2 = 138;\ 138 \times 2\frac{1}{2} = 345$$

Clearly, 46 should come in place of question mark.

Type II- To Find the Wrong Term

in this type of questions, a series is given in which one of term is required to detect the pattern of series and find that wrong term.

Ex. 35 In the following series, a wrong number is given. Find out that wrong number 4, 5, 10, 18, 34, 59, 95.
Sol. Series pattern

$4+1^2=5;\ 5+2^2=9\neq 10;\ 9+3^2=18, 18+4^2=34;\ 34+5^2=59;\ 59+6^2=95$

Clearly, 10 is the wrong number and will be replaced by 9.

Ex. 36 Find the wrong number in the given series 84, 83, 79, 70, 52, 29.
Sol. Series pattern

$84-1^2=83;\ 83-2^2=79;\ 79-3^2=70;\ 70-42=54\neq 52;\ 54-5^2=29$

Clearly, wrong number is 52.

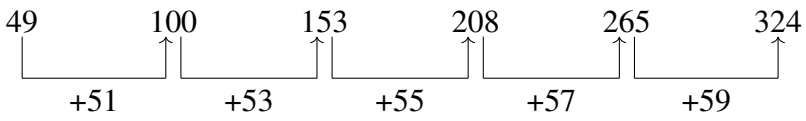
Type III- To Find the Missing Series Based on a Given Series

Two number series are given in this type question. The pattern of the second series is same as the pattern of the first series and on the basis of the first series, we have to find out the unknown terms of second series.

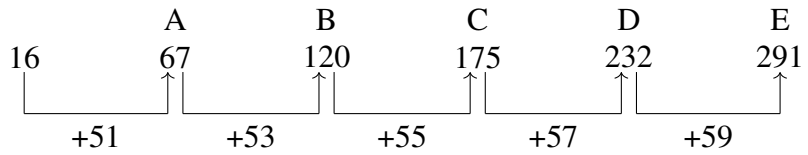
Ex. 37. There are two number series given below. First number series is arranged in a particular way and second series is based on first series. On the basis of this, which number will come at the place of D.

49	100	153	208	265	324
16	(A)	(B)	(C)	(D)	(E)

Sol.



Similarly,



So, the required number = 23 + 4 = 27